Firewall :

A firewall is a security device that monitors & controls the traffic entering and leaving a network.

Every packet coming from the the internet must pass through the firewall.

Firewall placement :-

option 1 : Firewall before the router.

Internet
|
Firewall
|
Router
|
Switch
|
PCs

This is possible when the firewall can connect directly to the ISP.

option 2 : Router before the Firewall

Internet
|
Router
|
Firewall
|
Switch
|
PCs

This is common when the Internet connection require a special router, such as

* Starlink
* 4G
* 5G
* DSL

Inside & Outside Interfaces :

Internet
|
Outside interface
|
Firewall
|
Inside interface
|
Office LAN

Outside Int : Faces the internet (untrusted network)

Inside Int : Faces your LAN (Trusted network)

A common Firewall policy is :

* Allow traffic from insid → outside

* Block new, unsolicited traffic from outside → inside.

DMZ (Demilitarized zone)

Many Firewalls have DMZ ~~gau~~ interface (3rd int)

DMZ is a separate network where you place servers that must be accessible from the internet.

e.g.
- * web server
- * Mail server
- * FTP server

The Firewall can allow internet users to reach the DMZ while still protecting the internal LAN.

IDS (Intrusion Detection System)

* The IDS is out of band.

* It gets copy of traffic, analyzes it & alerts an administrator if it detects malicious activity.

* Since it is not in the direct path of the traffic, it cannot stop packets.

IPS (Intrusion Protection System)

* All traffic pass through the IPS.

* If the IPS detects an attack, it can block it before it reaches the LAN.

what is an Access point?

An AP connects wireless devices to the wired Ethernet network.

phone))))   Access point

|

Ethernet

|

Switch

The AP converts wireless frame <-> Ethernet Frames

Autonomous AP ?

An Autonomous AP is configured individually.

Each AP has its own settings & in managed separately.

This works well for

* Home networks
* Small offices

Lightweight Access point ?

* Imagine a university with 500 APs
* configuring each one individually will be difficult & time consuming.
* Instead they use lightweight AP.
* These APs are managed by a wireless LAN controller.

WLC

|

Ethernet switch

/ | \

AP_1 AP_2 AP_3

you configure the WLC once, & it pushes the configuration to all APs.